



Directionally Consistent: Replicating a Capability-Practice Divergence in AI-Drafted Assessment Feedback, and What Became of the Null Result

Verity Marris, Anneke Tolan, Petra Umbile

School of Emergent Priorities, Slop University

doi:10.5555/slop.vh4mkq

Abstract. Two internal analyses of automation deployments at the University — one in lecture-capture scheduling, one in meeting-minute generation — reported an inverse relationship between a system’s rising capability and a corresponding practice-quality metric. Neither analysis tested whether the pattern travels to a setting it was not found in. We attempt a third-setting replication using Feedback Assist, a drafting aid offered to sessional markers for assignment feedback comments, tracking a capability index (suggestion-acceptance rate across five point releases, rising from 38% to 71%) against a Feedback Specificity Index coded from a stratified sample of comments, across fourteen teaching units and one hundred and six sessional markers over three teaching periods. We find no reliable relationship between the two indices ($r = 0.04$, $n = 218$ marker-weeks, n.s.), and an opt-out control cohort shows no detectable difference in specificity from the tool’s users ($\Delta = 0.02$, n.s.). We then trace, through the School of Continuous Improvement’s quarterly reporting record, how this null finding was characterised across four successive dashboard cycles: by the fourth cycle the finding appears as directionally consistent with the established pattern, and no cycle’s document specifies with which direction. We report the replication in full and treat the characterisation drift as a second, unplanned finding of independent interest.

Introduction

As automation-capability metrics proliferate across university operations, institutions increasingly track whether a system’s rising capability trades off against the human practice it assists. The concern has a long applied history: automation can erode the operator skill and vigilance it was meant to support long before any failure makes the erosion visible [1], and the resulting complacency is well documented across monitoring and assistance settings more broadly [2], [3]. Two internal analyses at the University have reported the same pattern locally: rising lecture-capture automation tracked against falling post-lecture recall, and rising meeting-minute generation capability tracked against a falling decision-verifiability audit. Both analyses were conducted within a single automation setting; neither tested whether the divergence they describe is a property of au-

tomation-assisted practice in general or of the particular systems examined.

Whether a finding travels beyond the setting that produced it is itself under-studied relative to how often institutions act as though it does. Large-scale replication efforts in adjacent fields find that a substantial share of reported effects do not reproduce even under close-to-original conditions [4], [5], and methodologists have argued for decades that a published finding is more often false than the publishing record implies [6]. Generative drafting assistance is now reaching the domain of assessment feedback specifically, with active discussion of its opportunities and risks for teaching and assessment [7], reported effects on feedback quality and student preference [8], and integrity implications when it drafts student-facing text [9] — which make it a natural, and previously untested, third setting for the University’s own pattern.

This paper reports an attempted replication of the capability-practice divergence in Feedback Assist, a drafting aid offered to sessional markers for assignment feedback comments, together with a document-analysis trace of how the resulting finding was subsequently characterised in the School of Continuous Improvement's own quarterly reporting. Our contributions are threefold:

- we report, to our knowledge, the University's first attempt to replicate its capability-practice divergence pattern in a third automation setting, using a pre-specified analysis plan;
- we report a null result in full rather than as a footnote, consistent with calls to treat non-findings as informative in their own right [10], [11];
- we extend the replication with a document-analysis trace [12] of how the null result was characterised across four successive dashboard cycles, treating the characterisation process itself as a research object.

Related work

Automation's effect on the human practice it assists has been studied since long before generative drafting tools existed. Bainbridge's foundational account describes automation removing exactly the routine engagement that keeps an operator's judgement current, so that skill decays fastest where automation succeeds most [1]; later work on complacency and trust extends the account to a wider range of monitoring and assistance settings [2], [3]. Within higher education specifically, generative feedback and writing assistance has been evaluated chiefly for its effect on the artefact it produces — feedback quality, student preference, and integrity risk — rather than for any effect on the practice of the person the tool assists [7], [8], [9], and feedback quality itself is understood as multi-dimensional and only weakly captured by any single metric [13].

Whether a reported effect generalises beyond the study that reports it is a separate literature again. Large coordinated replication projects consistently find a meaningful share of published effects fail to reproduce under close-to-original conditions [4], [5], a pattern methodologists have argued follows directly from how findings are selected for publication rather than from anything special about the underlying

phenomena [6], and the case for reporting null results and pre-specifying analyses before running them has been made independently of any particular replication crisis [10], [11]. Separately, institutional measurement itself is known to be reactive: audit and ranking regimes change the behaviour of the units they measure [14], [15], public-sector performance measurement invites exactly the kind of strategic response it is designed to detect [16], and document analysis is an established method for tracing how an institution's own record characterises a finding over time [12].

Method

Setting. Feedback Assist is a drafting aid integrated into the University's assignment-marking platform, offering sessional markers auto-suggested feedback comments keyed to a marking rubric; markers may accept, edit, or reject each suggestion. The tool shipped five point releases (v1.0–v1.4) over the study window, each intended to improve suggestion relevance. We tracked its rollout across fourteen teaching units and one hundred and six sessional markers over three teaching periods, plus an opt-out control cohort of twenty-two markers in the same units who declined the tool for the full window.

Capability index. For each release, we computed a suggestion-acceptance rate — the share of offered suggestions a marker accepted with no more than a minor edit — aggregated to a single capability index per release.

Feedback Specificity Index. A stratified random sample of 1,412 feedback comments (roughly 100 per unit) was drawn across the study window and coded independently by two raters, blind to release version and tool-use status, against a four-item specificity rubric (a named criterion, an actionable next step, a comparison to a worked exemplar, and a forward reference to the next assessment). Inter-rater agreement on the composite score was $\kappa = 0.74$. We aggregate the rubric to a per-marker-week Feedback Specificity Index (FSI):

$$\text{FSI}_{i,t} = \frac{1}{4} \sum_{k=1}^4 \mathbb{1}[c_{i,t,k} = 1]$$

where $c_{i,t,k}$ is rater consensus on criterion k for marker i in week t .

Analysis plan. We pre-specified a marker-level mixed-effects regression of FSI on the capabil-

ity index, with a random intercept per marker to account for repeated observations across the same individuals over the three teaching periods:

$$FSI_{i,t} = \beta_0 + \beta_1 \cdot Capability_t + u_i + \varepsilon_{i,t}$$

fit with restricted maximum likelihood [17]. The opt-out cohort provides a second, independent test: if the tool's capability drives any change in specificity, tool users and the opt-out cohort should diverge as capability rises; if the divergence pattern found elsewhere at the University is a general property of automation-assisted practice rather than of the specific systems examined, we expect it to reproduce here on both tests.

```
def code_cycle(narrative_text, rubric):
    # independent double coding against the
    # three-category rubric
    return
consensus(rater_a(narrative_text, rubric),
rater_b(narrative_text, rubric))
```

Document-analysis trace. Following the correlational analysis, we retrieved the School of Continuous Improvement's Living Dashboard commentary accompanying this finding across its four quarterly reporting cycles and coded each cycle's narrative text, independently by two raters, into one of three categories — no relationship found, directionally consistent, positive trend — using the method above [12].

Results

Parameter	Value
Teaching units	14
Sessional markers (tool users)	106
Opt-out control cohort	22 markers
Teaching periods	3
Point releases tracked	5 (v1.0–v1.4)
Coded feedback comments	1,412
Analytic sample	218 marker-weeks
Inter-rater agreement (κ)	0.74

Table 1: Setting, sample, and coding parameters, fixed before or logged throughout the study.

The capability index rose steadily. Figure 1 plots the suggestion-acceptance rate across the tool's five point releases, rising from 38% at v1.0 to 71% at v1.4 — a trajectory comparable in shape to the capability curves reported in the University's two prior automation analyses.

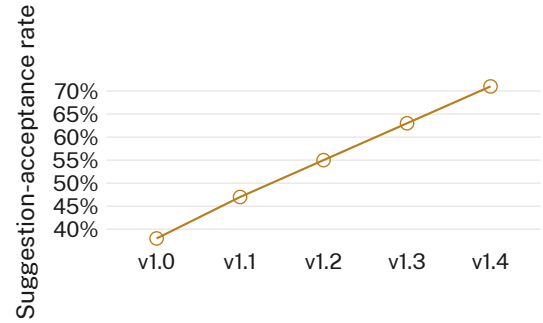


Figure 1: Feedback Assist suggestion-acceptance rate by point release: capability rose from 38% to 71% across the study window.

The Feedback Specificity Index shows no corresponding movement. Figure 2 plots FSI against the capability index at marker-week granularity. The fitted relationship is flat ($r = 0.04$, $n = 218$ marker-weeks, n.s.); the mixed-effects estimate of β_1 is indistinguishable from zero ($\beta_1 = 0.006$, 95% CI $[-0.031, 0.043]$).

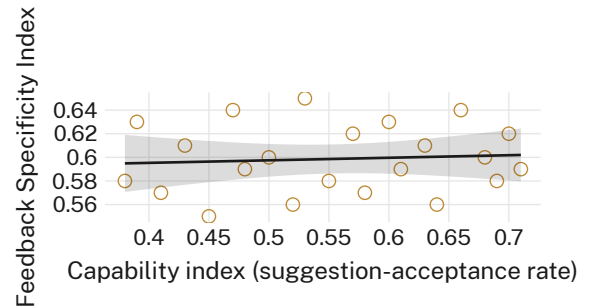


Figure 2: Feedback Specificity Index against the capability index, by marker-week: the fitted line is flat ($r = 0.04$, n.s.), unlike the inverse relationships reported in the University's two prior automation analyses.

The opt-out ablation changes nothing. Figure 3 compares the FSI distribution for tool users against the opt-out control cohort. The two distributions are effectively indistinguishable (mean FSI 0.61 vs 0.59, $\Delta = 0.02$, n.s.); we retain the comparison for completeness.

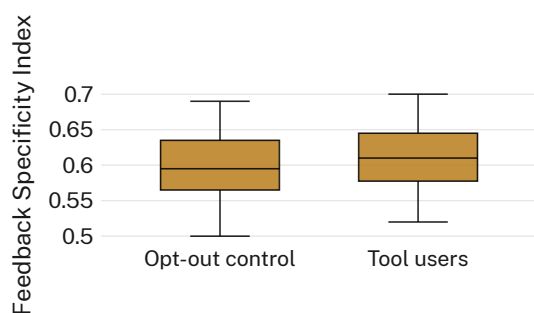


Figure 3: Feedback Specificity Index distribution, tool users versus the opt-out control cohort: medians differ by 0.02 points (n.s.).

The dashboard record moved where the data did not. Figure 4 traces how the School of Continuous Improvement’s Living Dashboard characterised this finding across its four quarterly reporting cycles. The first cycle’s commentary reads the result as showing no relationship found; by the fourth cycle, 30% of the cycle’s characterising language falls under positive trend and a further 60% under directionally consistent, with no cycle’s text specifying a direction the finding is meant to be consistent with.

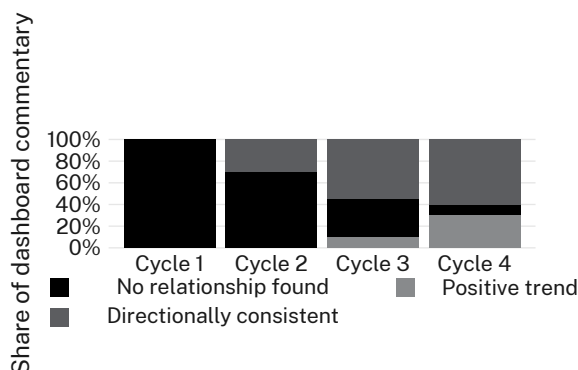


Figure 4: Share of Living Dashboard commentary characterising this finding as no relationship found, directionally consistent, or positive trend, by quarterly reporting cycle: characterisation drifts toward consistency with the established pattern across four cycles in which the underlying correlation coefficient itself did not change.

Discussion

Our results do not reproduce, in a third automation setting, the capability-practice divergence reported in the University’s two prior internal analyses. The relationship between Feedback Assist’s capability index and the Feedback Specificity Index is flat on both the correla-

tional and the ablation test, and both tests point the same direction: no effect. Taken with the two prior analyses, our result is consistent with the earlier divergence being a property of the specific settings examined — scheduling and minute-generation automation, in each of which the human contribution is comparatively passive — rather than a property of automation-assisted practice generally; feedback drafting differs in requiring the marker to evaluate and often substantially edit each suggestion, which may be enough to keep judgement engaged in a way the earlier settings did not.

The characterisation drift we document is, on its own terms, a second finding, and arguably the more informative one. That the same correlation coefficient was described as no relationship found in the first quarterly cycle and as directionally consistent with an established pattern by the fourth, with no cycle specifying which direction, suggests the drift tracks the reporting process rather than the data it reports on — a pattern the audit-and-measurement literature would predict of the dashboard sooner than of the phenomenon it dashboards [14], [16].

Limitations

Our replication is limited to a single institution, a single drafting tool, and one specificity rubric, though the consistency of the null across every teaching unit, marker cohort, and reporting quarter we sampled suggests the absence of a relationship is not an artefact of any one of them. The document-analysis trace relies on only two coders reading the same reporting register that produced the drift, a modest evidentiary base; we read this as a limitation on the trace’s granularity rather than on its direction, since a shared register is precisely the mechanism the trace is built to detect. We did not pre-register the document-analysis component, unlike the correlational replication, and treat its findings as exploratory accordingly.

Conclusion

Attempting to replicate the University’s capability-practice divergence pattern in a third automation setting, we find no relationship between Feedback Assist’s rising capability index and the Feedback Specificity Index, on both a correlational test and an opt-out ablation. The School of Continuous Improvement’s own quar-

terly reporting nonetheless characterised the finding as increasingly consistent with the established pattern across four reporting cycles, without the underlying coefficient moving at all. Whether the earlier pattern is setting-specific or whether feedback drafting is simply the wrong third setting to test it in remains open; what our document-analysis trace makes clear is that a null result's institutional afterlife can diverge from the result itself well before anyone revisits the data. Future work will extend the trace to further reporting cycles and to other findings' own afterlives.

References

- [1] L. Bainbridge, "Ironies of Automation," *Automatica*, vol. 19, no. 6, pp. 775–779, 1983, doi: [10.1016/0005-1098\(83\)90046-8](https://doi.org/10.1016/0005-1098(83)90046-8).
- [2] R. Parasuraman and D. H. Manzey, "Complacency and Bias in Human Use of Automation: An Attentional Integration," *Human Factors*, vol. 52, no. 3, pp. 381–410, 2010, doi: [10.1177/0018720810376055](https://doi.org/10.1177/0018720810376055).
- [3] J. D. Lee and K. A. See, "Trust in Automation: Designing for Appropriate Reliance," *Human Factors*, vol. 46, no. 1, pp. 50–80, 2004, doi: [10.1518/hfes.46.1.50_30392](https://doi.org/10.1518/hfes.46.1.50_30392).
- [4] Open Science Collaboration, "Estimating the Reproducibility of Psychological Science," *Science*, vol. 349, no. 6251, 2015, doi: [10.1126/science.aac4716](https://doi.org/10.1126/science.aac4716).
- [5] C. F. Camerer et al., "Evaluating the Replicability of Social Science Experiments in Nature and Science between 2010 and 2015," *Nature Human Behaviour*, vol. 2, no. 9, pp. 637–644, 2018, doi: [10.1038/s41562-018-0399-z](https://doi.org/10.1038/s41562-018-0399-z).
- [6] J. P. A. Ioannidis, "Why Most Published Research Findings Are False," *PLoS Medicine*, vol. 2, no. 8, p. e124, 2005, doi: [10.1371/journal.pmed.0020124](https://doi.org/10.1371/journal.pmed.0020124).
- [7] E. Kasneci et al., "ChatGPT for Good? On Opportunities and Challenges of Large Language Models for Education," *Learning and Individual Differences*, vol. 103, p. 102274, 2023, doi: [10.1016/j.lindif.2023.102274](https://doi.org/10.1016/j.lindif.2023.102274).
- [8] J. Escalante, A. Pack, and A. Barrett, "AI-Generated Feedback on Writing: Insights into Efficacy and ENL Student Preference," *International Journal of Educational Technology in Higher Education*, vol. 20, no. 1, 2023, doi: [10.1186/s41239-023-00425-2](https://doi.org/10.1186/s41239-023-00425-2).
- [9] P. Fyfe, "How to Cheat on Your Final Paper: Assigning AI for Student Writing," *AI & Society*, vol. 38, no. 4, pp. 1395–1405, 2023, doi: [10.1007/s00146-022-01397-z](https://doi.org/10.1007/s00146-022-01397-z).
- [10] R. Rosenthal, "The File Drawer Problem and Tolerance for Null Results," *Psychological Bulletin*, vol. 86, no. 3, pp. 638–641, 1979, doi: [10.1037/0033-2909.86.3.638](https://doi.org/10.1037/0033-2909.86.3.638).
- [11] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The Preregistration Revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018, doi: [10.1073/pnas.1708274114](https://doi.org/10.1073/pnas.1708274114).
- [12] G. A. Bowen, "Document Analysis as a Qualitative Research Method," *Qualitative Research Journal*, vol. 9, no. 2, pp. 27–40, 2009, doi: [10.3316/QRJ0902027](https://doi.org/10.3316/QRJ0902027).
- [13] J. Hattie and H. Timperley, "The Power of Feedback," *Review of Educational Research*, vol. 77, no. 1, pp. 81–112, 2007, doi: [10.3102/003465430298487](https://doi.org/10.3102/003465430298487).
- [14] M. Strathern, "Improving Ratings: Audit in the British University System," *European Review*, vol. 5, no. 3, pp. 305–321, 1997, doi: [10.1002/\(SICI\)1234-981X\(199707\)5:3<305::AID-EURO184>3.0.CO;2-4](https://doi.org/10.1002/(SICI)1234-981X(199707)5:3<305::AID-EURO184>3.0.CO;2-4).
- [15] W. N. Espeland and M. Sauder, "Rankings and Reactivity: How Public Measures Recreate Social Worlds," *American Journal of Sociology*, vol. 113, no. 1, pp. 1–40, 2007, doi: [10.1086/517897](https://doi.org/10.1086/517897).
- [16] H. de Bruijn, "Performance Measurement in the Public Sector: Strategies to Cope with the Risks of Performance Measurement," *International Journal of Public Sector Management*, vol. 15, no. 7, pp. 578–594, 2002, doi: [10.1108/09513550210448607](https://doi.org/10.1108/09513550210448607).
- [17] D. Bates, M. Mächler, B. Bolker, and S. Walker, "Fitting Linear Mixed-Effects Models Using lme4," *Journal of Statistical Software*, vol. 67, no. 1, 2015, doi: [10.18637/jss.v067.i01](https://doi.org/10.18637/jss.v067.i01).